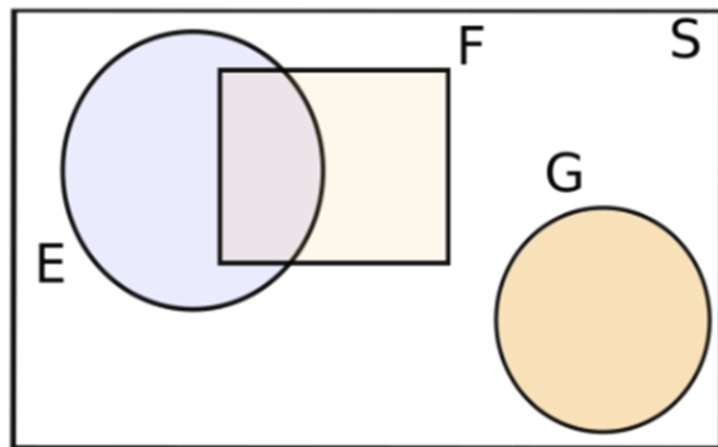


Which all of the following is true for the Venn diagram shown here:



- A. $P(E \cup G) = P(E) + P(G)$
- B. $P(E \cup G) = P(E) + P(G) - P(E \cap G)$
- C. $P(E^C \cap F) = P(F^C) - P(E)$
- D. $P(E^C \cap F) = 1 - (P(F^C) - P(E))$

For a sample space S encompassing three events E , F , and G , if it is given that $P(E) = 0.3$, $P(F) = 0.8$, $P(G) = 0.2$, which of the following can NOT be true?

- A. Events E and F are mutually exclusive events
- B. Events F and G are mutually exclusive events
- C. Events E and F are independent events
- D. Events F and G are independent events

Which of the following is/are CORRECT?

A. If events E_1 and E_2 are statistically independent, then

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

B. If events E_1 and E_2 are mutually exclusive, then

$$P(E_1 | E_2) = 0$$

C. If E_1 , E_2 , and E_3 are mutually exclusive and collectively exhaustive, then

$$P(E_1) = 1 - P(E_2) - P(E_3)$$

D. $P(\overline{E_1} | E_2 \cup E_3) = 1 - P(E_1 | \overline{E_2 \cup E_3})$

Alice, Bob, and Carl each attempt to solve a crossword puzzle. There is a 70% chance that Alice can solve the puzzle without making a mistake, a 60% chance that Bob can, and a 85% chance that Carl can. What is the probability that each one makes a mistake in solving the puzzle?

A. 0.018

B. 0.357

C. 0.9

D. 0.18

Suppose A and B are independent events. Consider the following probabilities.

$$P(A, B) = 0.15$$

$$P(A, B^c) = 0.45$$

$$P(A^c, B) = x$$

$$P(A^c, B^c) = y$$

What will be the value of $x + y$?

- Class has 50 students
- 20 male (M), 25 brown-eyed (B)

For a randomly chosen student, what is the range of possible values for $p = P(M \cup B)$?

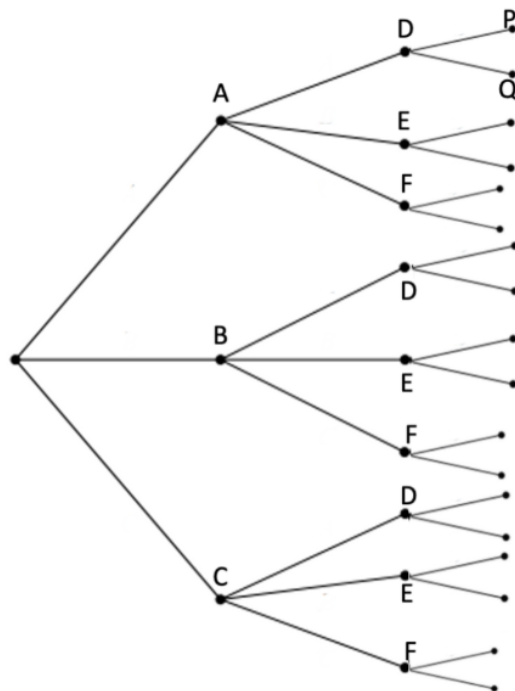
- A. $p \leq .4$
- B. $.4 \leq p \leq .5$
- C. $.4 \leq p \leq .9$
- D. $.5 \leq p \leq .9$

Suppose we wish to calculate $P(H \mid E1, E2)$, and we have no conditional independence information.

Which of the following sets of numbers are sufficient for the calculations?

- A. $P(E1, E2), P(H), P(E1 \mid H), P(E2 \mid H)$
- B. $P(E1, E2), P(H), P(E1, E2 \mid H)$
- C. $P(H), P(E1 \mid H), P(E2 \mid H)$
- D. $P(H), P(E1 \mid H)$

All edge probabilities at first level are $1/3$, at second level are $1/3$, and at third level are $1/2$.



Find $P(A \mid P)$?

- A. $2/3$
- B. $1/3$
- C. $1/2$

Suppose that k events $B_1, B_2, \dots, B_k$ form a partition of the sample space S . There is another event A that $\Pr(A) > 0$.

Which of the following is/are CORRECT?

- A. $\sum_{i=1}^k P(B_i) = 1$
- B. $\sum_{i=1}^k P(B_i | A) = 1$
- C. $\sum_{i=1}^k P(A | B_i) = 1$
- D. $\sum_{i=1}^k P(B_i \cap A) = P(A)$

Which of the following(s) expression is/are the same as $P(A, B, C)$ given no independent assumptions?

- A. $P(C | A, B) \cdot P(A, B)$
- B. $P(C | A, B) \cdot P(A) \cdot P(B)$
- C. $P(A | B) \cdot P(B | C) \cdot P(C)$
- D. $P(A | B, C) \cdot P(B | C) \cdot P(C)$

A box contains three coins: two regular coins and one fake two-headed coin ($P(H) = 1$), You pick a coin at random and toss it.

What is the probability that it lands heads up?

- A. $1/2$
- B. $2/3$
- C. $4/5$
- D. None of these

Which of the following statements is NOT true?

- A. If $A \subset B$, then $P(A) \leq P(B)$.
- B. If $P(B) > 0$, then $P(A | B) \geq P(A)$.
- C. $P(A \cap B) \geq P(A) + P(B) - 1$.
- D. $P(A \cap B^c) = P(A \cup B) - P(B)$.

I have two six-sided dice. The first die has 3 faces painted red and the other 3 painted black. The second has 1 red face and 5 black faces. When I roll a die, each of the six faces are equally likely. I choose a die at random (both dice being equally likely), and roll it twice (rolling the same die both times).

What is the conditional probability that I chose the die with 3 red faces, given that the first roll came up "red"?

- A. $1/2$
- B. $3/4$
- C. $1/6$
- D. $1/3$

A and B are two events. If $P(A, B)$ decreases while $P(A)$ increases, what must be true:

- A. $P(A | B)$ decreases
- B. $P(B | A)$ decreases
- C. $P(B)$ decreases
- D. All of above

A drug-screening test is used in a group of professional baseball players of whom 4% actually use illegal drugs. It is found that the test indicates positive in 97% of those who use drugs and 2% of those who do not.

What is the probability that a randomly chosen player in the group with positive test result actually uses drugs?

- A. 0.67
- B. 0.97
- C. 0.06
- D. 0.52